

Question ID: 79be5180

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Medium

Question

$y = 9x + 12$

$x + 7y = 20$

The solution to the given system of equations is (x, y) . What is the value of y ?

Correct Answer: 3

Rationale

The correct answer is **3**. It's given that $y = 9x + 12$. Substituting $9x + 12$ for y in the second equation in the system, $x + 7y = 20$, yields $x + 7(9x + 12) = 20$, which gives $x + 63x + 84 = 20$, or $64x + 84 = 20$. Subtracting **84** from each side of this equation yields $64x = -64$. Dividing each side of this equation by **64** yields $x = -1$. Substituting -1 for x in the first equation in the system, $y = 9x + 12$, yields $y = 9(-1) + 12$, or $y = 3$. Therefore, the value of y is **3**.